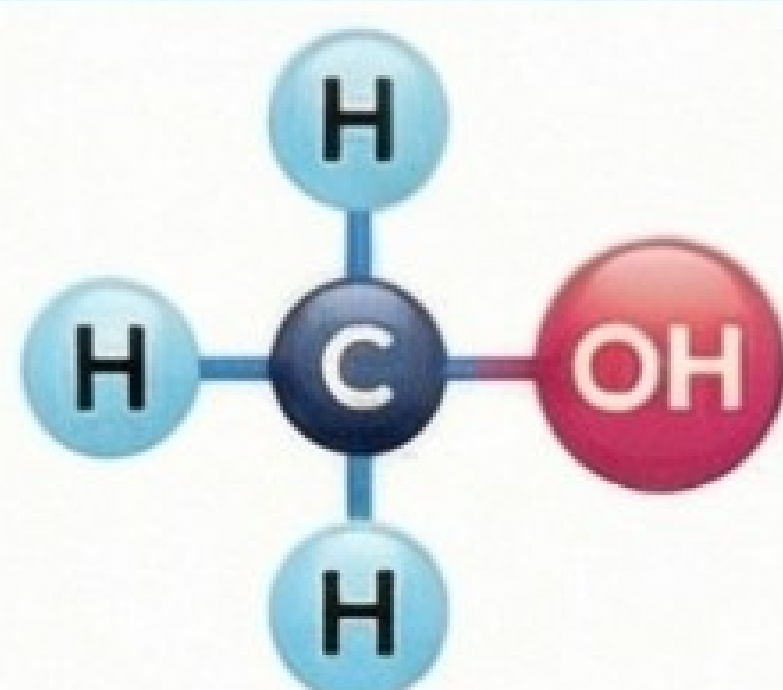


A Visual Guide to Alcohols, Phenols & Ethers

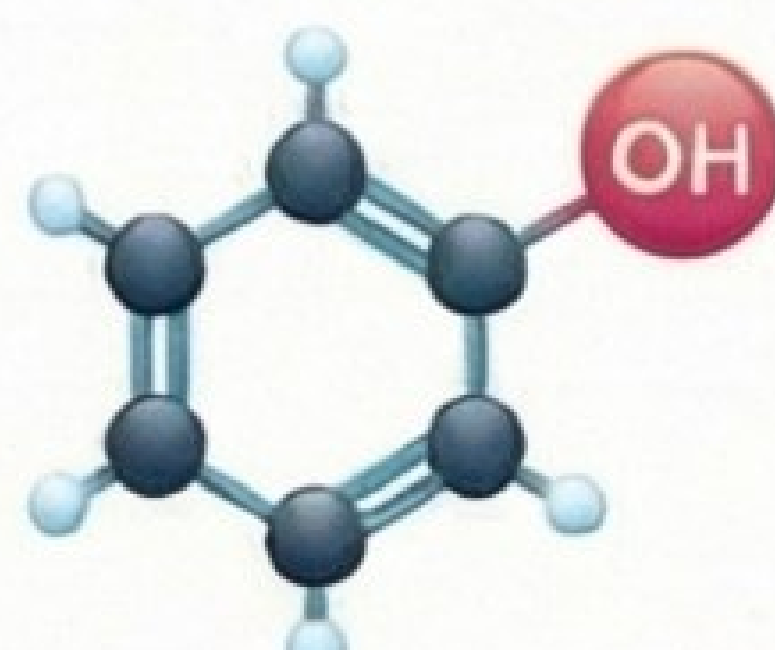
1. Meet the Functional Groups

Alcohol (R-OH)



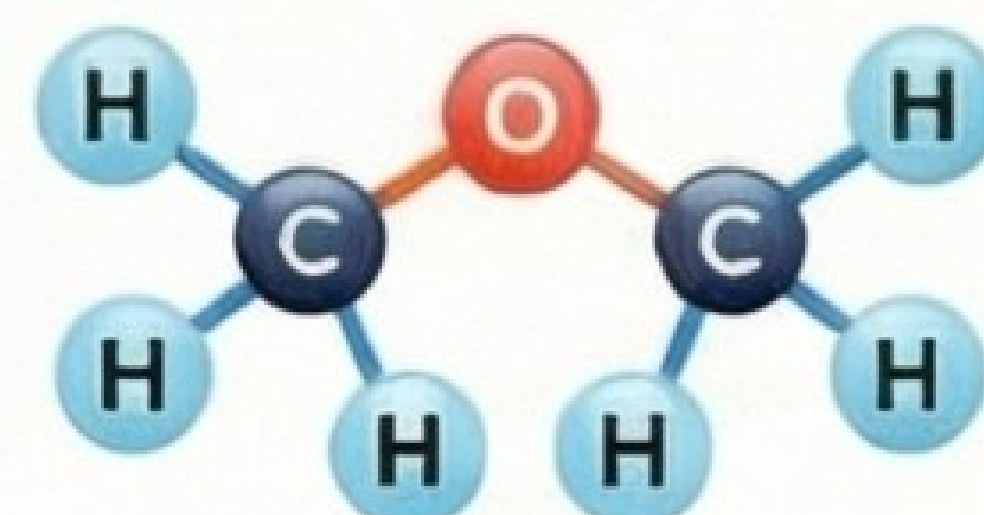
An -OH group is attached to a saturated carbon atom (part of an alkyl group, R).
Example: $\text{CH}_3\text{-OH}$ (Methanol).

Phenol (Ar-OH)



An -OH group is directly attached to a carbon atom within a benzene ring.
Example: Benzene ring with an -OH group.

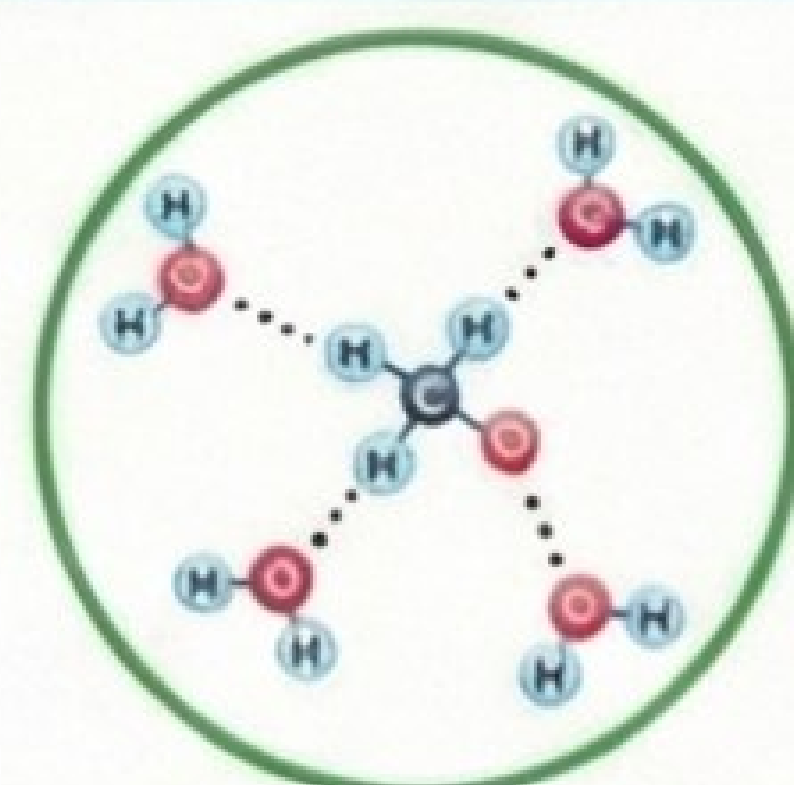
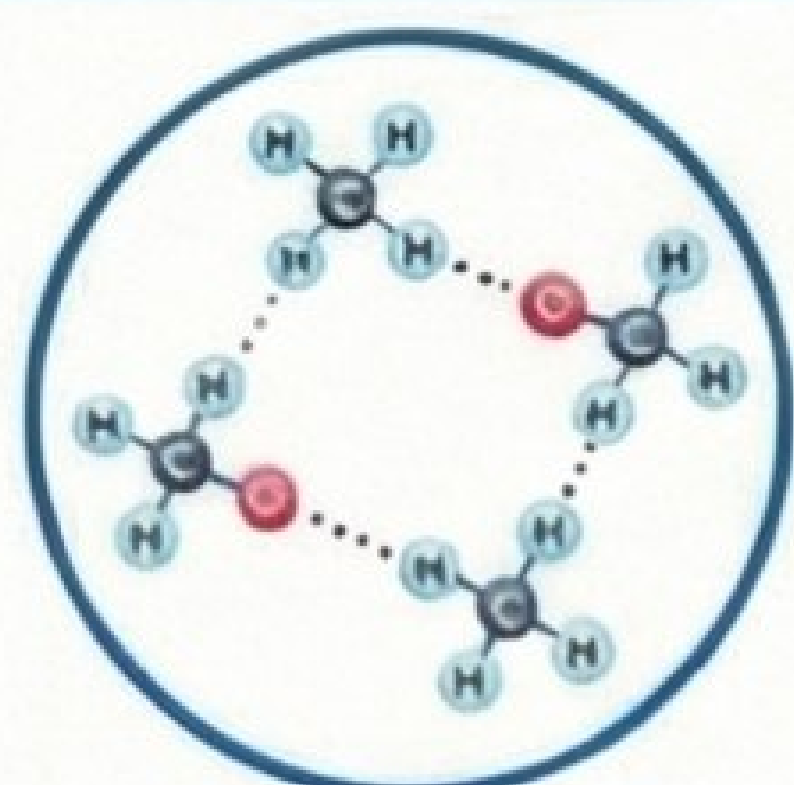
Ether (R-O-R')



An oxygen atom is bonded to two alkyl or aryl groups.
Example: $\text{CH}_3\text{-O-CH}_3$ (Dimethyl ether).

2. The World of Alcohols

Properties



High Boiling Points

Due to intermolecular hydrogen bonding between -OH groups. Boiling point decreases with increased branching.

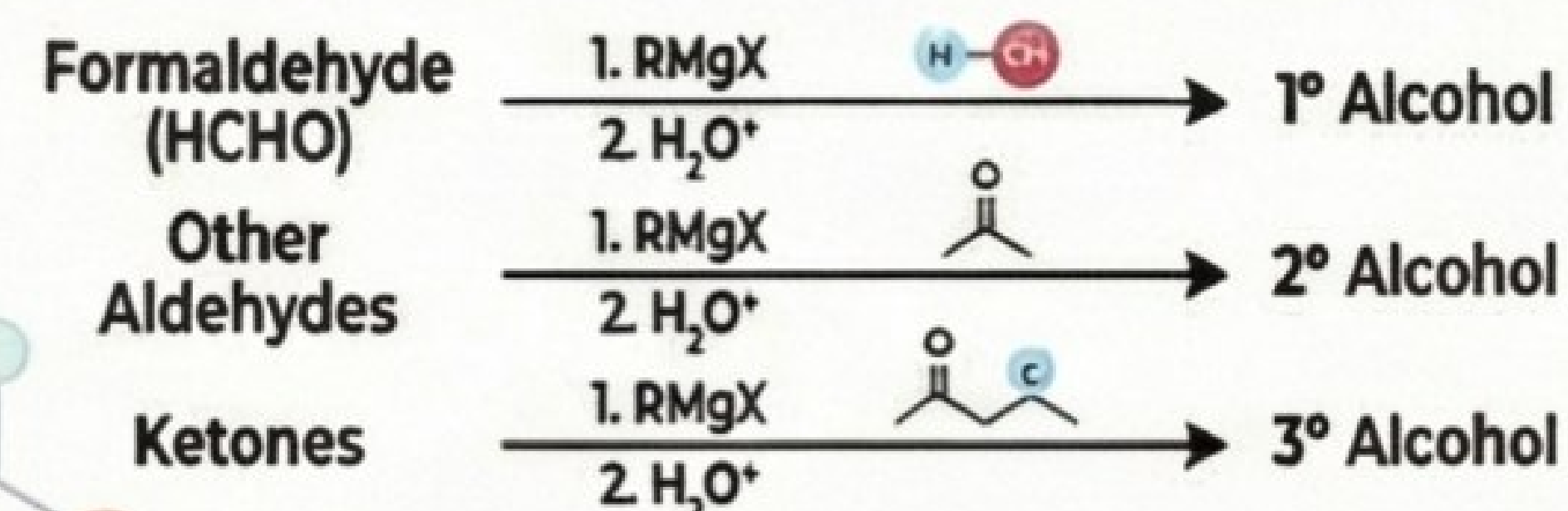
Soluble in Water

Smaller alcohols form hydrogen bonds with water molecules.

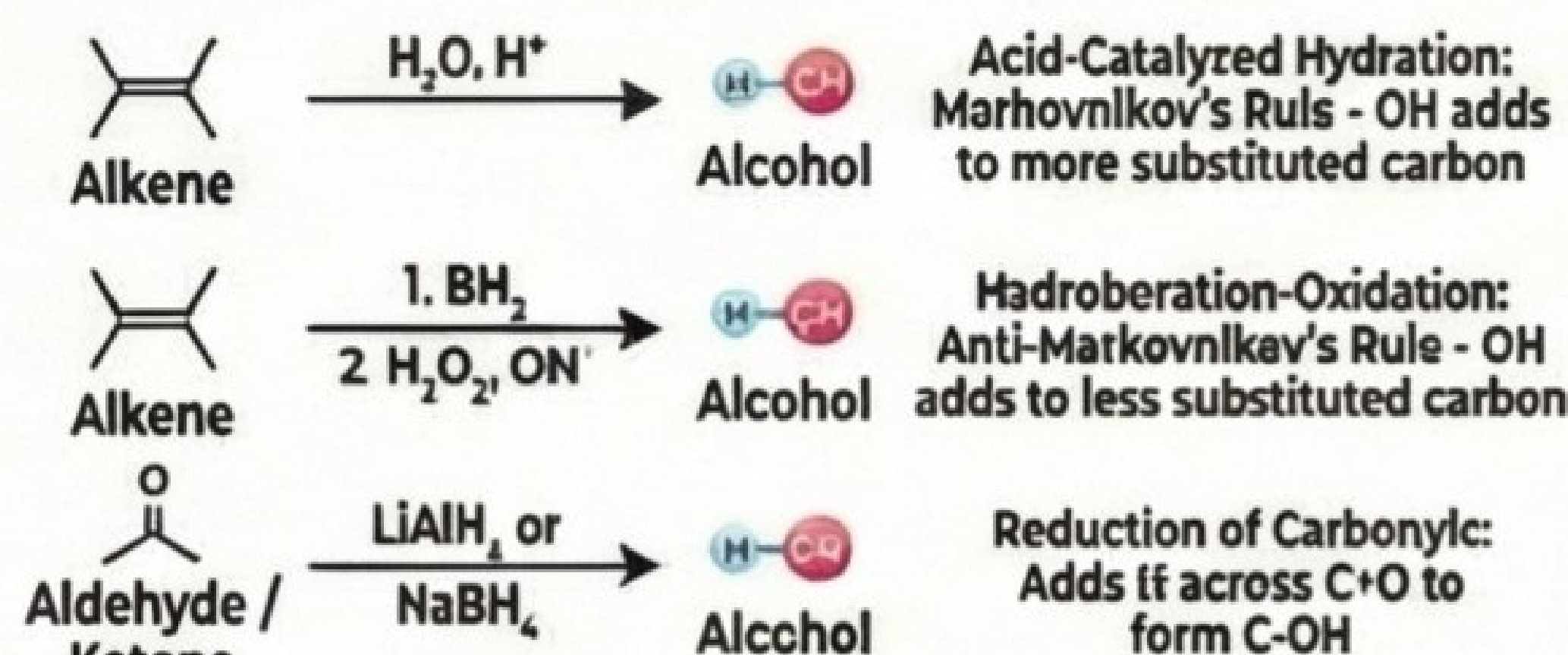
Acidity

Phenol > Water > Alcohol

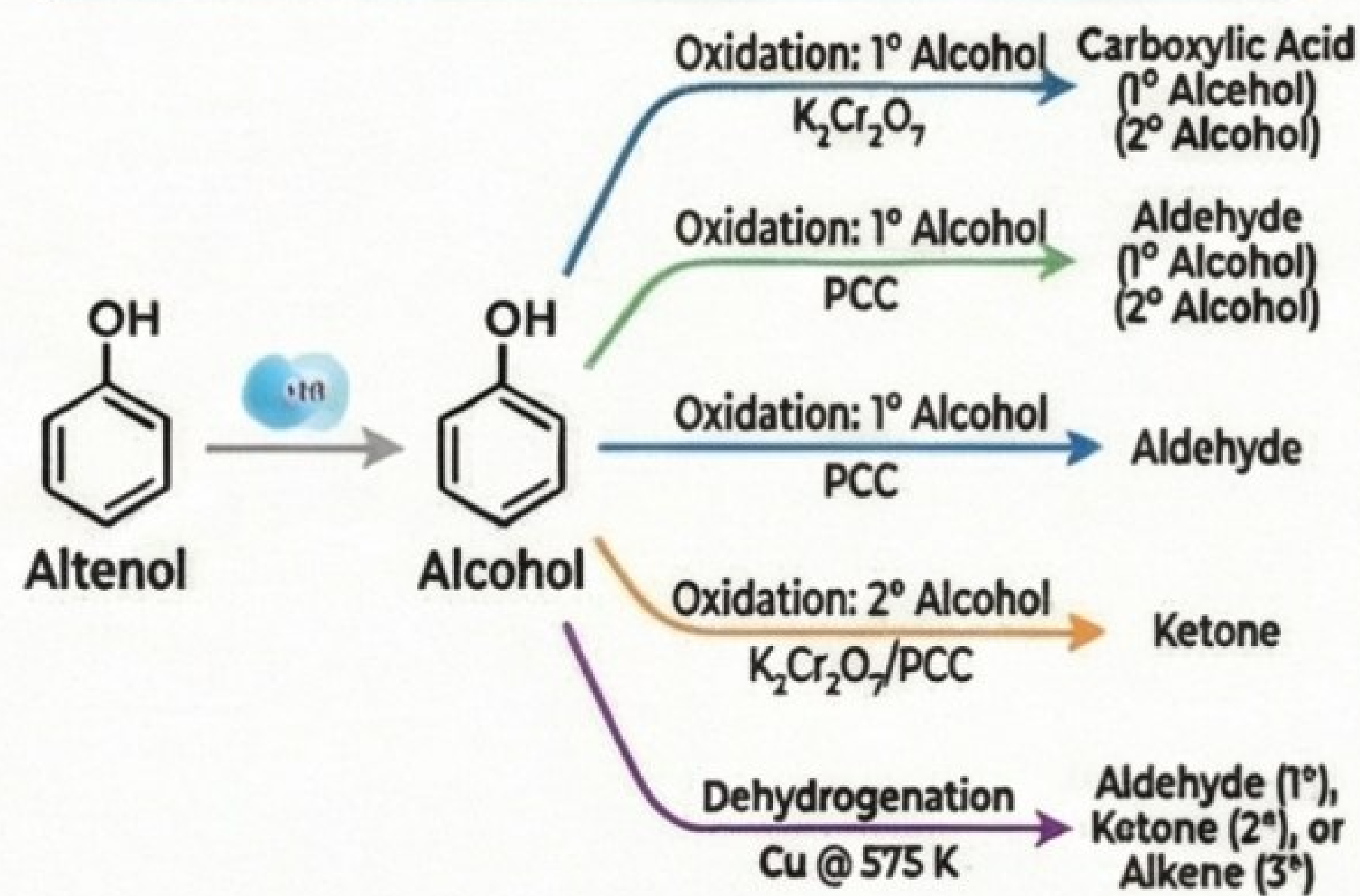
Alcohols are less acidic than phenols and water, partly due to the +I effect of alkyl groups.



Key Preparation Methods

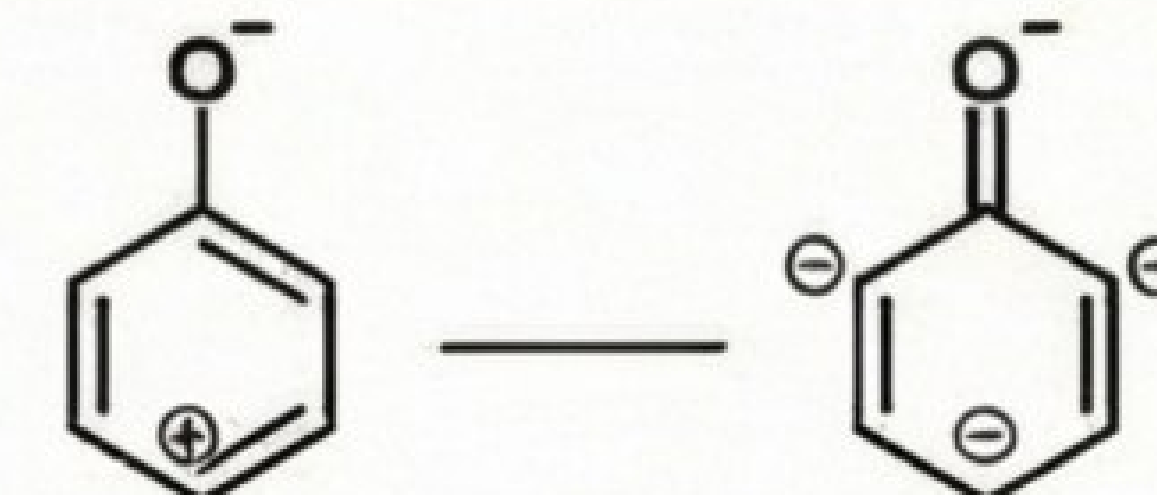


Key Reactions of Alcohols



3. The Special Case of Phenols

Significantly More Acidic than Alcohols



Phenoxide ion is highly stable due to resonance delocalization.

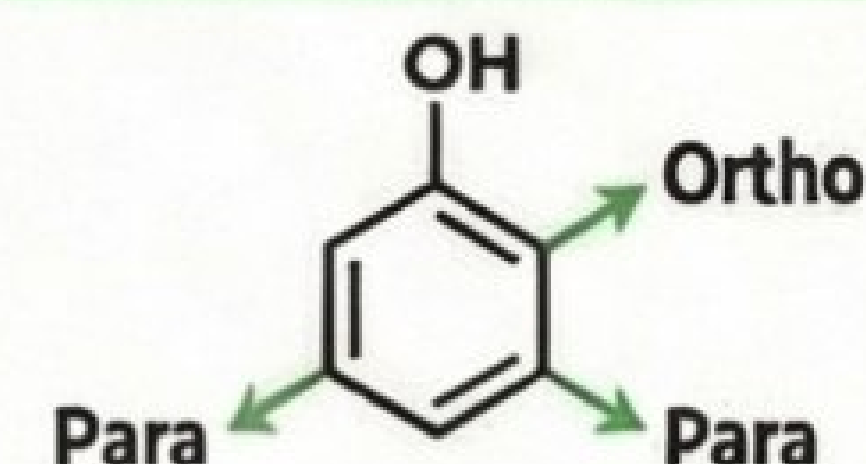
EWG (e.g., -NO₂)
Increase Acidity

Pull electron density, stabilize phenoxide.
Order: Para > Ortho > Meta.

EDG (e.g., -CH₃)
Decrease Acidity

Push electron density, destabilize phenoxide.

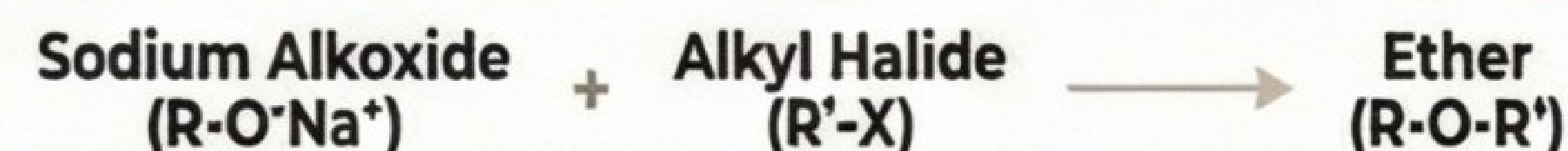
Directs Incoming Groups



Activates ring for electrophilic substitution, primarily at Ortho & Para due to +R effect.

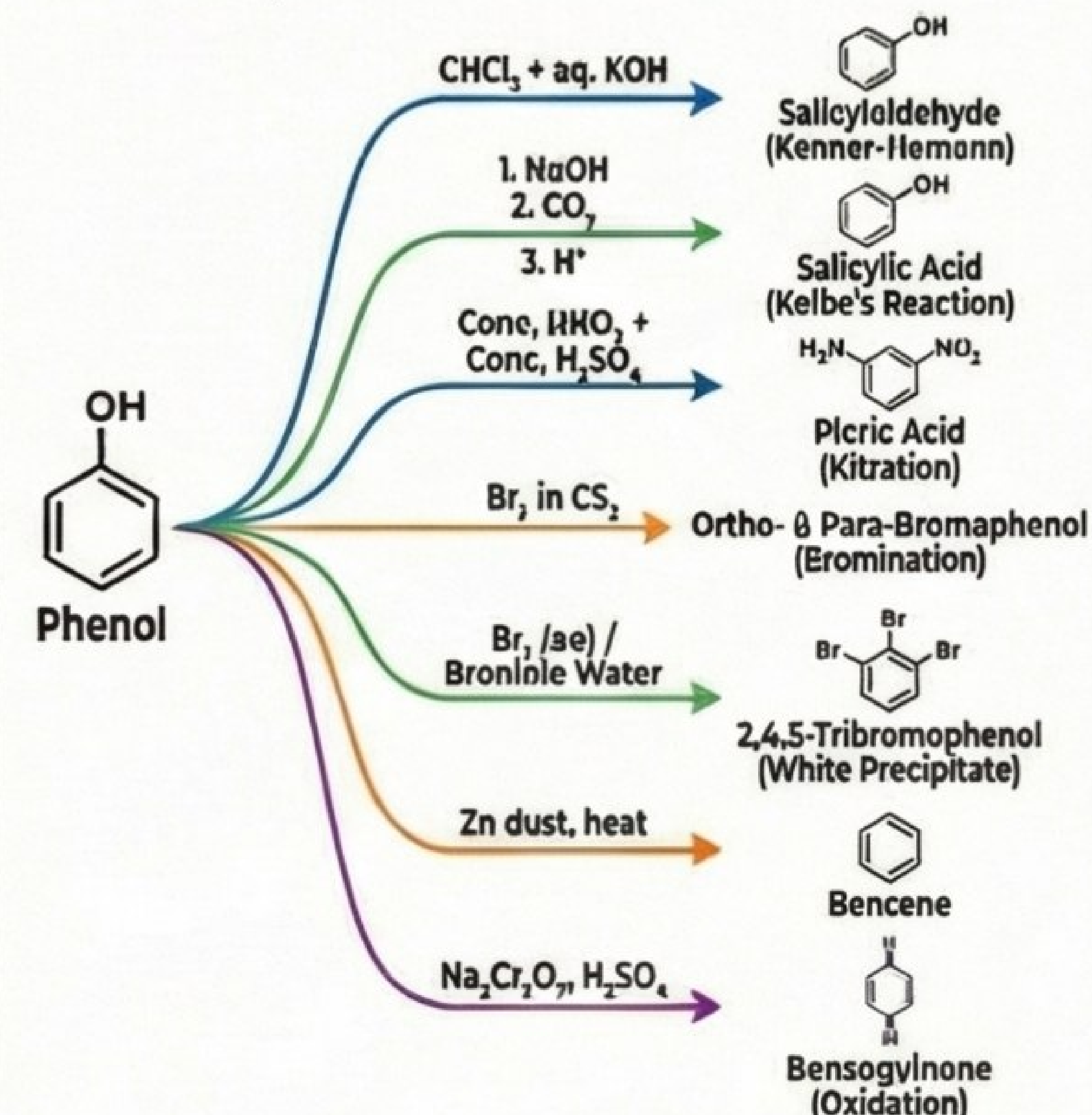
4. Understanding Ethers

Premier Preparation: Williamson Synthesis

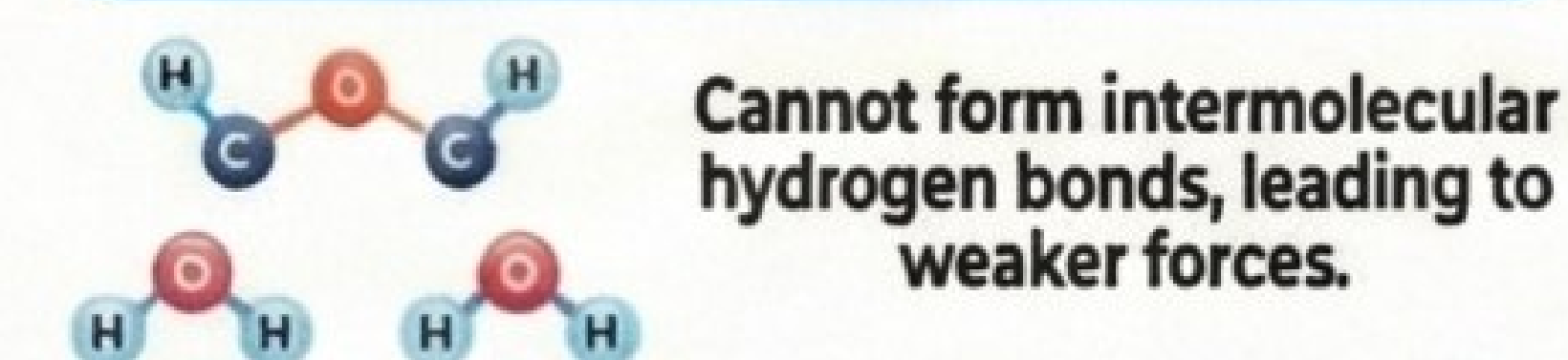


Rule: Big Alkoxide, Small Halide for best results.

Important Reactions of Phenol

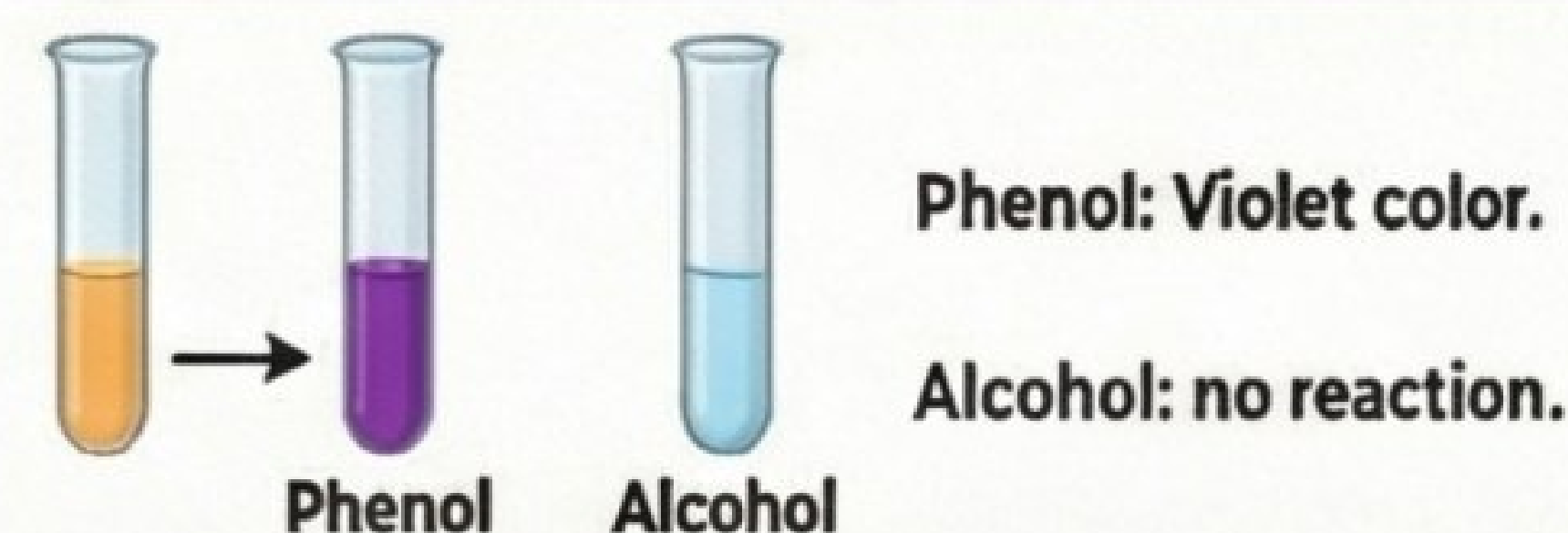


Lower Boiling Points



5. How to Tell Them Apart: Chemical Tests

Test 1: Neutral FeCl₃ Test



Test 2: Bromine Water Test

